

## A CONCRETE QUESTION ABOUT FINITE OBSERVERS

# Finite Directional Records as a Concrete Observer Model

Can one localized system acquire, retain, and expose a relational orientation record while its information, support, energy cost, and gravitational footprint are controlled in one semiclassical regime?

## The observer-level question

Quantum-gravity observer rules assign finite operational content to an observer subsystem, while de Sitter observer algebras include the observer in the construction [1]-[4]. We ask whether a narrow physical register can make some assumptions auditable: one localized system records the relative orientation of another, stores it for a stated proper time, and permits later relational readout.

The aim is not to identify a rotor with an observer, but to expose which information, dynamics, localization, and gravitational inputs an observer rule needs.

## Why direction is useful

Finite spatial reference frames and their degradation are established [5]-[7]. We use a wholly relational task: if  $O$  is the register and  $D$  the target, the unknown is their alignment, not orientation relative to coordinates:

$$g_{rel} = g_0^{(-1)} g_D \text{ in } SO(3). \quad (1)$$

Performance is the Haar-prior global risk for estimating  $g_{rel}$ . It probes the full rotational orbit, unlike local sensitivity or Hilbert-space size alone.

## The record variable

For the covariant orientation ensemble, we begin with the relative entropy of rotational asymmetry:

$$S_{dir} := D(\rho_0 \parallel G_{SO(3)}[\rho_0]) = S(G_{SO(3)}[\rho_0]) - S(\rho_0). \quad (2)$$

$S_{dir}$  is relative entropy of rotational asymmetry. It is not identified with classical memory dimension, thermodynamic entropy, Hilbert-space dimension, Casimir, quantum Fisher information, or Harlow's  $S_{Ob}$ . Whether comparison with  $S_{Ob}$  is meaningful is review question 1.

Lifetime is operational: after time  $T$ , a permitted relational readout must still attain the declared risk. This is classical-record stability, not generic phase coherence.

## The proposed theorem chain

Fix a class  $C_{dir}$  of localized register-target models, states, environments, actions, and preparation/storage/readout protocols. The intended theorem quantifies over this class:

$$\begin{aligned} R_{rel}(T) &\geq F_{record}(S_{dir}, \text{degradation}, \text{channel error}) \\ S_{dir} &\leq F_{capacity}(\text{energy}, \text{support}, \text{gravity}; \text{model data}) \\ \Rightarrow \Phi(R_{rel}(T), T, \text{support}, \text{gravity}; \text{model data}) &\geq 0. \end{aligned} \quad (3)$$

Rigorous arrows exist on named domains: global risk requires rotational resources; confined orbital matter converts them into support and excitation-energy cost; a stipulated isotropic heat channel gives a conditional degradation law.

## The primary bottleneck

The common-action physical record channel is **open**. One local action must derive acquisition, storage degradation, and readout; bound the microscopic KMS-to-effective-channel error; and enter the same stress-energy ledger. Existing de Sitter master equations specify the needed approximation, not that it follows from thermality [10].

## What gravity and the realization add

On fixed de Sitter, exact conserved-source, transmission, and electric-Weyl maps define a gauge-invariant exterior witness [11]. No rigorous completed-source interval yet excludes zero. Even after zero exclusion, this is not a capacity bound; collapse, horizon or quantum-extremal-surface displacement, or another invariant may be relevant.

A supported massive Skymion supplies one certified matter candidate, alongside curved and gravitating de Sitter Skymions [12]. It establishes only that the supported-profile subproblem is nonempty; the architecture must survive rejection of this wall, membrane, or field model.

## Why request feedback now

The project is at a fork before its most expensive bridge. We seek direction on the  $S_{dir}$ -to- $S_{Ob}$  relation, the gravitational quantity constraining a finite register, and whether a one-action semiclassical model clarifies the observer rule without starting from a closed-universe path integral. We seek criticism and redirection, not endorsement or a publication decision.

**Claim boundary:**  $S_{dir} \neq S_{Ob}$ ; record stability is not quantum coherence; measurement back-action is not gravitational backreaction; fixed-background response is not self-consistent gravity.

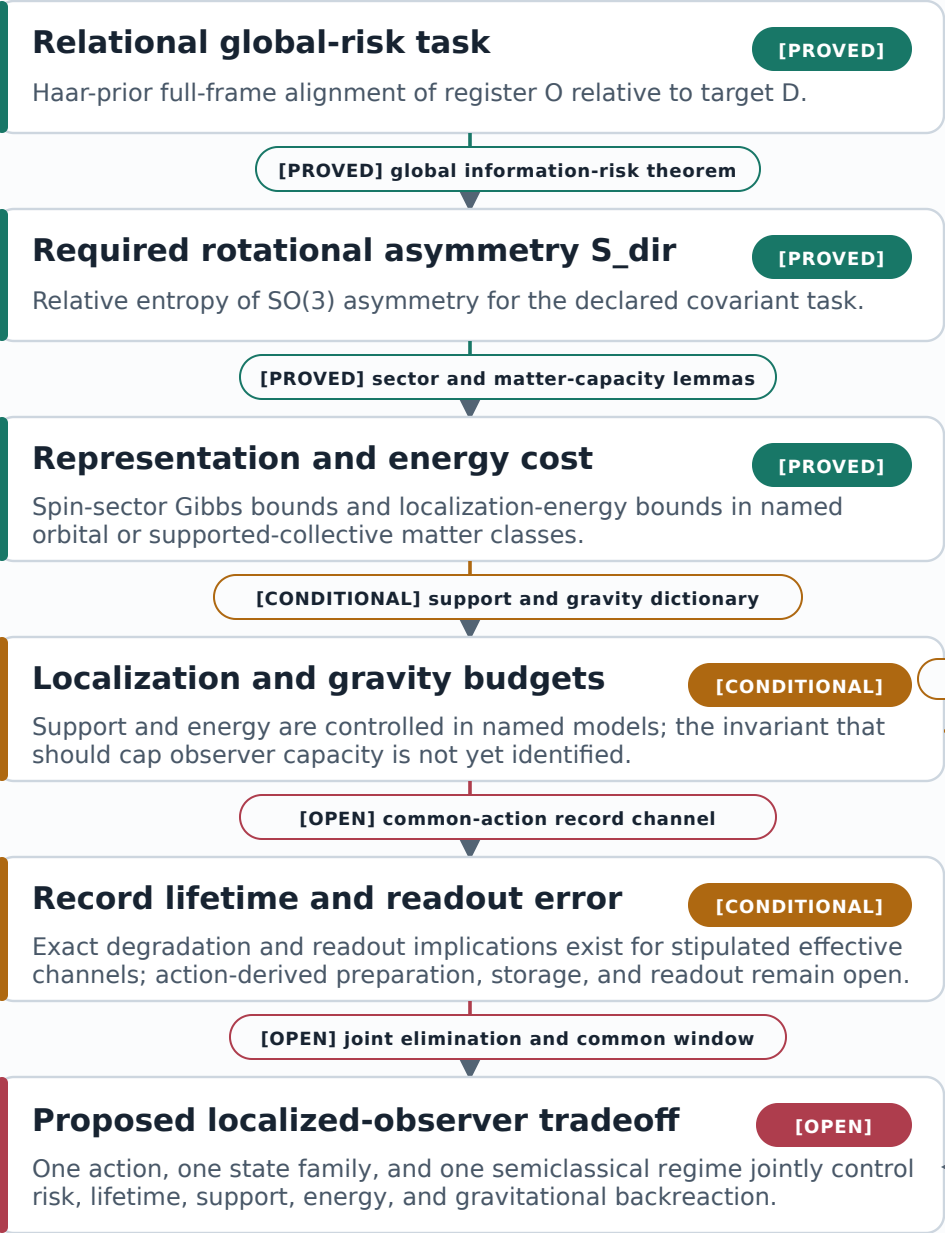
# Finite directional records as an observer model

Dependency map: rigorous ingredients, conditional bridges, and the theorem still to be built

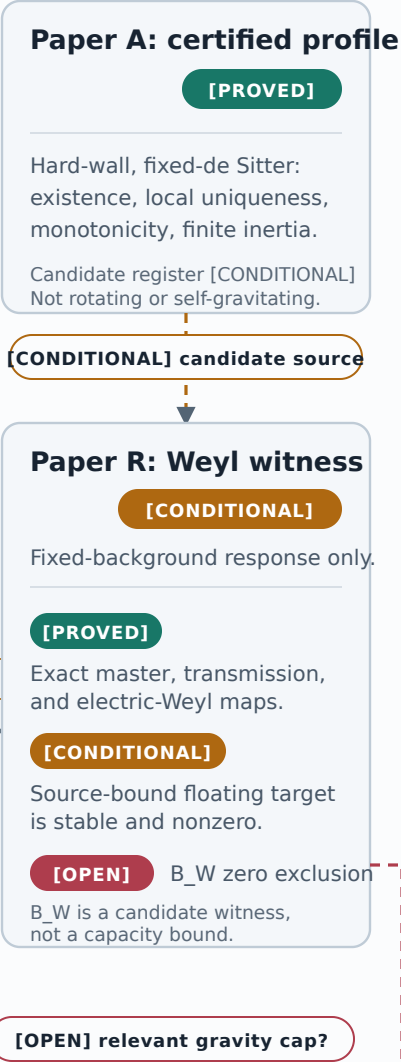
Task: infer the relative  $SO(3)$  alignment of two localized physical systems - never an absolute direction.

**STATUS**    **[PROVED]** theorem on a declared domain    **[CONDITIONAL]** model-dependent or floating evidence    **[OPEN]**

## OBSERVER-REGISTER CHAIN



## REALIZATION SUPPORT



### NON-NEGOTIABLE CLAIM BOUNDARIES

**[PROVED] Relational, not absolute**  
The datum is alignment between two physical systems,  
not orientation relative to coordinates.

**[OPEN]  $S_{dir} \neq S_{Ob}$**   
No observer-entropy dictionary is proved.

**[CONDITIONAL] Fixed background**  
 $B_W$  is a candidate witness, not the capacity bound  
and not self-consistent gravity.

**[OPEN] Common-action channel**  
Preparation, storage, and readout are not yet derived together.

## THE PURPOSE OF THIS PACKET

# Where the observer model should connect to quantum gravity

These questions are intentionally upstream of the remaining interval calculation. A negative answer to any one of them would redirect the architecture while that redirection is still inexpensive.

- 1 Is  $S_{\text{dir}}$ , used here as an operational  $S(3)$ -asymmetry resource for a covariant alignment task, meaningfully comparable to the observer entropy  $S_{\text{ob}}$  in the observer rule, or are these fundamentally different resources?
- 2 Which gauge-invariant gravitational quantity should constrain the capacity of a finite observer register in a de Sitter causal patch: collapse, horizon or quantum-extremal-surface displacement, exterior response, or something else?
- 3 Would a one-action semiclassical model deriving relational record formation, lifetime, localization, readout, and gravitational footprint clarify the observer rule, or is a closed-universe gravitational path integral essential from the outset?

**Feedback target.** We are not asking whether the present realization is complete. We are asking whether this is the right operational bridge to build, which quantity should replace the provisional gravitational witness if needed, and which observer-level identification should remain deliberately absent.

# Auditable Status of the Directional-Record Program

**[PROVED]** means an analytic or computer-assisted theorem on the stated domain. **[CONDITIONAL]** means an exact implication under a stipulated effective model or source-bound floating evidence. **[OPEN]** means a missing implication, norm estimate, or physical identification. A composite claim inherits the least-complete label of its ingredients.

## A. Relational orientation as the operational task

**[PROVED]** Let the unknown relative orientation  $g$  in  $S_0(3)$  have normalized Haar prior, let an arbitrary POVM return  $g_{\text{hat}}$ , and use full-frame chordal cost  $\sin^2(\theta(g_{\text{hat}}^{-1}g)/2)$ . The reference representation may contain arbitrary rotation-trivial multiplicities. For every normal state and POVM, including mixed states and invariant ancillas,

$$\begin{aligned} R_{\text{ref}} &\geq 1/(16 \langle J^2 \rangle + 8), \\ R_{\text{ref}} &\geq c_{S_03} \exp[-2 S_{\text{dir}}/3], \quad c_{S_03} = 6/(e \pi^{5/3}), \\ S_{\text{dir}} &= A_{S_03}(\rho) = D(\rho \parallel G_{S_03}(\rho)). \end{aligned} \tag{A.1}$$

The first bound follows from the spin-1 fusion matrix and a discrete Hardy inequality. The second follows from relative-entropy data processing for the covariant orbit ensemble and a global rate-distortion bound. Neither proof assumes a locally unbiased estimator, Cramer-Rao regime, purity, polarization, fixed irrep, or finite-dimensional multiplicity. This is Research Theorem W3 in THEOREMS.md and docs/global\_so3\_reference\_risk.md.

**[PROVED]** The datum is the alignment of two physical frames, never orientation relative to coordinates. The loss is global over the group, not a local-sensitivity proxy. For strict integer-spin support through  $j=J$ ,  $R_{\text{ref}} \geq \sin^2[\pi/(2J+3)]$ . The fermionic Finkelstein-Rubinstein sector has its separately stated projective bound.

**[PROVED]** Here  $S_{\text{dir}}$  is only relative entropy of rotational asymmetry. Equation (A.1) gives the necessary allocation  $S_{\text{dir}} \geq (3/2) \log(c_{S_03}/R_{\text{ref}})$  where positive. It does not establish accessible classical memory, sufficiency, or a relation to  $S_{0b}$ .

**[CONDITIONAL]** If the reference alone undergoes stipulated isotropic rotational heat diffusion with exposure  $\Gamma(T) = \int_0^T \gamma(t) dt$ , the spin-1 score is attenuated exactly:

$$\begin{aligned} R_{\text{ref}}(T) &\geq (3/4) [1 - \exp(-2 \Gamma)] \\ &\quad + \exp(-2 \Gamma) / (16 \langle J^2 \rangle + 8). \end{aligned} \tag{A.2}$$

This is a finite-time record-risk theorem for that effective channel, not preservation of arbitrary quantum coherence. **[OPEN]** No one-action protocol yet controls preparation, storage, readout, microscopic-to-effective channel error, support stress, and backreaction on one nonempty open parameter set.

## B. Capacity from localization and energy

**[PROVED]** For finitely many spinless nonrelativistic particles of total mass  $M$ , hard support  $|x_i| \leq a$ , and a ground-subtracted excitation Hamiltonian satisfying the quadratic-form inequality  $H_{\text{ex}} \geq \sum_i p_i^2/(2m_i)$ , every form-domain state obeys

$$\begin{aligned} \langle L^2 \rangle &\leq 2 M a^2 E_{\text{ex}}, \\ R_{\text{ref}} &\geq 1/(32 M a^2 E_{\text{ex}} + 8). \end{aligned} \tag{B.1}$$

The result includes mixed, zero-mean, and rare-tail states with rotation-trivial internal labels. It excludes intrinsic spin, relativistic or massless fields, soft localization, negative interaction energy, and nontrivially rotating light species. A separate compactness substitution is a declared proxy, not a general-relativistic body theorem. See W3a and docs/localized\_orbital\_reference.md.

**[PROVED]** More generally, sector floors  $\epsilon_j$  yield  $S_{\text{dir}} \leq \beta E + \log Z_H(\beta)$  only when  $Z_H(\beta) = \sum_j (2j+1)^2 \exp(-\beta \epsilon_j)$  is finite. Covariance plus finite mean energy is insufficient; a bounded invariant counterexample permits increasingly accurate frame states at uniformly bounded energy. A physical theorem must derive growing sector floors from its matter model.

**[PROVED]** In the hard-supported massive-Skyrme hedgehog collective family on fixed de Sitter, with support radius  $a$  and minimum support lapse  $N_w > 0$ , the static energy and collective inertia obey

$$\begin{aligned} I[F] &\leq 4 M[F] a^2/(3N_w), \\ E_j &\geq \sqrt{3N_w} j(j+1)/2/a. \end{aligned} \tag{B.2}$$

This produces a linear large-spin floor and finite integer or projective rotational partition function. It is uniform over radial relaxation inside the adiabatic collective family (W3b).

**[CONDITIONAL]** The Skyrmion is therefore a concrete candidate for the capacity premise, not a complete rotating-field observer. Collective-band completeness, noncollective modes, projection error, right/isospin access, and the capacity of a marked rotating wall remain uncontrolled.

## C. Certified supported profile

**[PROVED]** At the declared fixed-background hard-wall parameters, AU.1 is an exact-rational computer-assisted local existence theorem. It proves one solution in a certified Newton ball, strict monotonicity, a negative wall slope, and finite positive rotor inertia. The source artifact is experiments/skyrmion\_newton\_reduced\_hessian\_rounded\_exact\_certificate.json, SHA256 prefix c4c95db47470.

Digest strings on pages 5-7 show the first 12 hexadecimal characters. Full SHA256 values and exact artifact paths are indexed in claim\_evidence\_ledger.md.

**[PROVED]** This is local uniqueness on fixed pure de Sitter with a prescribed Dirichlet wall. It is not global uniqueness, dynamical stability, a rotating solution, arbitrary wall dynamics, or Einstein-Skyrme backreaction. **[CONDITIONAL]** It establishes that the candidate used in later calculations is more than a shooting profile; the wall remains a physical modeling input, not a harmless regulator.

## D. Conserved source, master field, and electric Weyl

**[PROVED]** A static quadrupolar source cannot be specified by energy density alone. W3n gives the exact static even-parity bulk conservation equations and wall tractions. W3o shows that the undeformed rigid rotational Skymion stress fails this gate. This rejects the rigid source, not rotating Skymions: the centrifugal matter deformation and moving support must be included.

**[PROVED]** For the declared fixed-background matter-plus-moving-membrane weak form, W3z.15-W3z.16 establish a closed positive Friedrichs operator with  $||A^{-1}|| \leq 100$ , nonzero regular forcing, and a unique nonzero weak centrifugal deformation. The source artifacts have SHA256 prefixes 4a4e3ecd48a2 and 9edc2bc47953. Nonzero matter deformation alone does not imply nonzero exterior response.

**[CONDITIONAL]** Exact Hilbert-variation and shell identities, with source-bound mesh refinement, support a completed bulk-plus-moving-membrane stress at the default point. The analytic factorization is exact; closure of the selected branch is floating rather than interval certified. The safe wording is "source-bound floating conservation audit," not an unqualified certified source.

**[PROVED]** Once a valid conserved stress is supplied, the source, propagation, and readout maps are exact in one frozen convention. On fixed pure de Sitter, for the static  $\ell=2$  normalization, regular center behavior, and no-log horizon condition,

$$\begin{aligned} A_2 \Psi &= F, & A_2 &= -d/dr[(1-r^2/R^2)d/dr] + 6/r^2, \\ F &= 8\pi G[-r^2 N \rho'/6 + r(1+4r^2/R^2)\rho/6 \\ &\quad - r p_r/2 - j + 2r p_i], & \Delta E_{rr} &= -6 \Psi(r) Y_2/r^3. \end{aligned} \tag{D.1}$$

The Friedrichs operator has a positive exact Green kernel and  $A_2 \geq 6/R^2$ . W3t, W3v, and W3x prove the stress-to-master identity, ideal-shell transmission, and exterior master-to-electric-Weyl reconstruction. Their artifact prefixes are 3be17612134b, c83536779d2a, and 3025d65fe825. Exact maps do not certify a nonzero completed input amplitude.

**[CONDITIONAL]** Source-bound floating calculations reduce the exterior response to one scalar amplitude and give a stable nonzero target in the declared normalization. The feasibility artifact prefix is 584a22ea3ae9. This is design evidence, not a continuum interval including profile, origin, bulk, and wall errors.

**Witness boundary.** If nonzero, the proposed  $B_W$  would witness non-invisibility on a fixed background. It is not the observer-capacity bound, a self-consistent geometry, or a substitute for collapse, horizon/QES displacement, or full junction data.

## E. What remains in the response certificate

**[PROVED]** The certification architecture is exact. Outside the compact source, the fixed-background response factorizes into one amplitude  $A_{\text{ext}}=J_{\text{rigid}}+B(y)$ . If the symmetric weak form obeys  $q(y,v)=\ell(v)$  and the adjoint obeys  $q(v,z)=B(v)$ , arbitrary conforming trials satisfy

$$\begin{aligned} J_{\text{hat}} &= J_{\text{rigid}} + B(y_h) + R_y(z_h), \\ |A_{\text{ext}} - J_{\text{hat}}| &\leq \|R_y\|_{\ell^2}(q^*) + \|R_z\|_{\ell^2}(q^*). \end{aligned} \quad (\text{E.1})$$

Zero exclusion therefore requires the distance of the corrected-amplitude interval from zero to exceed the product of the full primal and adjoint dual-residual bounds. This is W3z.17; its physical premises are not yet all closed.

**[PROVED]** The current interval program certifies a positive-radius primal residual-square bound, an origin primal bound, exact internal conormal cancellation, the moving-wall adjoint load, and the positive-radius weak adjoint bulk load. The corresponding artifact hashes are:

- 814da74d5c21 (correlated outer primal residual),
- 7924fb7da3bb (origin primal residual),
- 1ee677788edc (conormal interface),
- ee73b3527750 (wall master load), and
- 3db6d390d521 (positive-radius adjoint bulk load).

These are certified proof components, not a response interval.

**[PROVED]** W3z.29 gives the current direct positive-radius-plus-wall adjoint form-dual estimate

$$\begin{aligned} \delta_{z,\text{partial}}^2 &\leq 0.592007476516919181, \\ \delta_{z,\text{partial}} &\leq 0.769420221021594403. \end{aligned} \quad (\text{E.2})$$

More than 99 percent of this bound comes from the adjusted bulk value coefficient, concentrated near the first positive-radius cells; the wall term is small. The exact artifact prefix is 500e56b5aa36.

**[OPEN]** The regular-origin master load is absent, so (E.2) is not a full  $\delta_{z,\text{partial}}$ . Even before that term is added, the conservative partial bound is too broad for zero exclusion. This quantified interval loss is a representation diagnostic, not a physical no-go.

**[OPEN]** No rigorous interval for the completed normalized electric-Weyl amplitude  $B_w$  presently excludes zero. The exact maps, favorable floating response, and unfinished physical zero-exclusion theorem are three separate facts.



## F. External claim boundary

**[PROVED]** The defensible mathematical core is: global relational  $S0(3)$  risk requires asymmetry and rotational resources; named localized matter classes convert rotational capacity into energy/support cost; one hard-wall profile and its fixed-background centrifugal inverse exist; and a valid conserved quadrupolar source has exact master, transmission, and Weyl maps.

**[CONDITIONAL]** The defensible physical construction is narrower: a hard-supported Skyrmion is a plausible semiclassical realization, a stipulated heat channel gives an exact degradation law, and source-bound floating calculations favor a nonzero fixed-background exterior response.

**[OPEN]** The project has not identified  $S_{dir}$  with  $S_{0b}$ ; derived acquisition-through-readout from one action; proved a common parameter window; interval-certified  $|B_W|>0$ ; completed tensorial Israel matching; solved a rotating self-gravitating Einstein-Skyrme-de Sitter geometry; or shown that exterior Weyl response caps observer information. A closed-universe path integral is outside the present claims.

**[OPEN]** Novelty, priority, venue fit, and publishability are not mathematical outputs of the repository. The separate literature matrix gives the primary-source comparison. Search absence is not used as proof of priority.

## G. Reproducibility map

| Status        | Result                                   | Primary repository pointer                                                               |
|---------------|------------------------------------------|------------------------------------------------------------------------------------------|
| [PROVED]      | Global risk/asymmetry/Casimir bounds     | THEOREMS.md W3; docs/global_so3_reference_risk.md; focused tests                         |
| [PROVED]      | Confined orbital capacity                | W3a; docs/localized_orbital_reference.md; focused tests                                  |
| [CONDITIONAL] | Heat-capacity composition                | W3a.1; experiments/universal_observer_tradeoff_certificate.json                          |
| [PROVED]      | Supported collective sector floor        | W3b; docs/supported_skyrmion_collective_spectral_floor.md                                |
| [PROVED]      | Validated hard-wall profile              | AU.1; exact Newton certificate listed on page 5                                          |
| [PROVED]      | Fixed-background source/master/Weyl maps | W3t, W3v, W3x; artifact hashes on page 6                                                 |
| [CONDITIONAL] | Floating completed response              | W3u, W3z.17; feasibility hash on page 6                                                  |
| [PROVED]      | Certified residual components            | W3z.24-W3z.29; hashes on page 7                                                          |
| [OPEN]        | Full Weyl zero exclusion                 | Needs full primal and adjoint form-dual residuals, including regular-origin adjoint load |

**[PROVED]** The complete wording guardrail, exact artifact index, and 219-test focused verification record are in docs/harlow\_review\_packet/claim\_evidence\_ledger.md. The full source comparison and its unresolved search limits are in literature\_matrix.md and source\_verification\_log.md.

**Adversarial distinctions.** Absolute orientation versus relational alignment; quantum coherence versus classical record stability;  $S_{dir}$  versus  $S_{0b}$ ; measurement back-action versus gravitational backreaction; and fixed-background response versus self-consistent gravity are audited separately.



## SHORT BIBLIOGRAPHY

# Conceptual Neighbors

These twelve primary sources anchor the review question. The separate literature matrix audits 33 sources across observer rules, finite reference frames, clocks, de Sitter open systems, gauge-invariant response, Skyrmons, and rigorous numerics.

1. D. Harlow, M. Usatyuk, and Y. Zhao, "Quantum mechanics and observers for gravity in a closed universe," [arXiv:2501.02359](#) (2025).
2. D. Harlow, "Observers, alpha-parameters, and the Hartle-Hawking state," [arXiv:2602.03835](#) (2026).
3. V. Chandrasekaran, R. Longo, G. Penington, and E. Witten, "An Algebra of Observables for de Sitter Space," [arXiv:2206.10780](#) (2022).
4. J. De Vuyst, S. Eccles, P. A. Hoehn, and J. Kirklin, "Gravitational entropy is observer-dependent," [arXiv:2405.00114](#) (2024).
5. E. Bagan, M. Baig, and R. Munoz-Tapia, "Aligning Reference Frames Using Quantum States," [arXiv:quant-ph/0106014](#) (2001).
6. S. D. Bartlett, T. Rudolph, R. W. Spekkens, and P. S. Turner, "Degradation of a quantum reference frame," [arXiv:quant-ph/0602069](#) (2006).
7. G. Gour, I. Marvian, and R. W. Spekkens, "Measuring the quality of a quantum reference frame: the relative entropy of frameness," [arXiv:0901.0943](#) (2009).
8. B. A. Costa and Y. E. Sanchez Sanchez, "An Upper Bound for the Mass of Microscopic Clocks," [arXiv:2602.24177](#) (2026).
9. L.-Q. Chen and F. Giacomini, "Quantum Reference Fields Transformations in Linearized Quantum Gravity," [arXiv:2606.09344](#) (2026).
10. M. Fukuma, Y. Sakatani, and S. Sugishita, "Master equation for the Unruh-DeWitt detector and the universal relaxation time in de Sitter space," [arXiv:1305.0256](#) (2014).
11. H. Kodama and A. Ishibashi, "A master equation for gravitational perturbations of maximally symmetric black holes in higher dimensions," [arXiv:hep-th/0305147](#) (2003).
12. Y. Brihaye and T. Delsate, "Skyrmion and Skyrme-Black holes in de Sitter spacetime," [arXiv:hep-th/0512339](#) (2006).

**Literature boundary.** Existing work already covers finite observer entropy, observer-dependent gravitational algebras, degrading quantum reference frames, physical clock limits, material reference fields, de Sitter open-system dynamics, gauge-invariant response, and curved/cavity Skyrmons. The proposed distinction is their quantitative conjunction in one narrow relational  $S(3)$  record model. No priority claim is made. Journal databases and terminology outside the targeted search were not exhausted.